SMART EMERGENCY GREEN CORRIDOR SYSTEM

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1. ABSTRACT

The Smart Emergency Green Corridor System is a proposed technology-driven solution designed to help reduce emergency response delays in congested urban areas. It connects ambulance location and route information with authorized traffic-signal coordination, hospital pre-arrival alerts, and a governed emergency-contact assistance workflow. Using GPS telemetry, route estimation, secure application services, and role-based dashboards, the system aims to improve coordination among ambulance operators, traffic command centres, and hospital emergency departments.

The project is intended to be developed in stages, beginning with a simulation-based prototype and progressing to a supervised pilot only after technical, safety, privacy, and operational approvals. Performance figures are treated as targets for evaluation rather than verified results.

2. INTRODUCTION

Urban traffic congestion can delay ambulances at junctions and along busy roads. Sirens and warning lights alert nearby road users, but they do not always create a clear path through dense traffic. Meanwhile, hospitals may receive limited advance information about the incoming patient, and responders may have difficulty identifying an unconscious patient or locating an emergency contact.

The Smart Emergency Green Corridor System proposes a coordinated workflow from ambulance dispatch to hospital arrival. The initial concept is intended for high-density Indian urban settings, including routes similar to major corridors in Chennai. Actual deployment locations should be selected through traffic analysis and consultation with the relevant authorities.

2.1 Problem Statement

Emergency response can be slowed by three connected gaps: (1) limited coordination between ambulance routes and traffic signals, (2) insufficient pre-arrival information for hospital emergency teams, and (3) difficulty finding verified emergency contacts when a patient cannot identify themselves. Separate systems may address individual parts of this chain, but the proposed project focuses on coordinating them through one governed platform.



Figure 1. Heavy urban congestion can obstruct emergency vehicle movement.

2.2 Why an integrated system?

- Traffic priority needs to be coordinated safely across multiple intersections.
- Hospital pre-alerts can give teams more preparation time when the data is timely and reliable.
- Emergency-contact assistance must use authorized sources and protect personal information.
- The system must include human oversight, failure handling, and safe fallback behaviour.

3. OBJECTIVES AND SCOPE

3.1 Objectives

- Develop an ambulance application for route guidance, location updates, and emergency request status.
- Estimate ambulance arrival times and identify upcoming intersections along an approved route.
- Request traffic-signal priority only through authorized interfaces and approved safety rules.
- Notify the receiving hospital of the ambulance ETA and permitted patient information.
- Provide a controlled workflow for emergency-contact assistance using approved data sources.
- Measure prototype reliability, usability, safety, and potential operational benefits.

3.2 Scope

- Role-based dashboards for traffic command, hospital emergency teams, and ambulance operators.
- GPS-based route and ETA support, with simulated traffic integration in the first prototype.
- Hospital pre-arrival alerts and destination updates.
- Emergency-contact workflow using synthetic data in the prototype until lawful access is approved.
- Audit logging, security controls, fallback handling, and an evaluation plan.

4. EXISTING SYSTEM AND LIMITATIONS

The source concept compares a reader-based RFID approach with GPS and cloud coordination. RFID may work well for local identification, but its usefulness for advance route coordination depends on reader placement and coverage. GPS-based coordination can provide route context over a wider area, but it also depends on device accuracy, mobile connectivity, map quality, and secure system design.

Dimension	Possible limitation in isolated systems	Proposed direction
Coverage	Local readers may cover only selected points.	Use GPS route context and configurable approach zones.
Route coordination	A single detection point may not support multiple junctions.	Estimate arrival times for upcoming intersections.
Hospital readiness	Traffic hardware may not alert the ER.	Send authenticated pre-arrival notifications.
Identity assistance	Not usually included in signal hardware.	Separate, authorized contact-assistance workflow.
Security	Depends on actual implementation and configuration.	Strong authentication, signed requests, access controls and audit logs.

Note: Neither RFID nor GPS/cloud systems are automatically secure or reliable in every situation. A fair comparison must use the actual equipment and procedures at the proposed pilot site.

5. SYSTEM OVERVIEW AND ARCHITECTURE

The platform is organized into user interfaces, coordination services, integration adapters, and governed data services. This modular design allows the prototype to be tested without immediately connecting to live traffic controllers or real patient records.

Layer / module	Main responsibilities
Ambulance mobile app	Secure login, location updates, route display, destination selection and request status.
Traffic command dashboard	View active requests, monitor signal acknowledgements, handle exceptions and apply authorized overrides.
Hospital emergency dashboard	View incoming ambulance ETA, acknowledge pre-alerts and prepare resources using approved information.
Coordination service	Validate requests, estimate route/ETA, manage state changes and coordinate notifications.
Integration adapters	Connect to approved traffic controllers, dispatch systems and hospital interfaces.
Security and audit layer	Authentication, role-based permissions, encryption, monitoring and event logging.

5.1 End-to-end workflow

Step	Workflow action
1	Ambulance crew or dispatch opens an emergency request.
2	System verifies user, vehicle, request and permissions.
3	GPS and route services estimate arrival times and upcoming intersections.
4	Traffic engine sends an authorized signal-priority request and waits for acknowledgement.
5	Hospital receives ETA and approved pre-arrival details.
6	Vehicle passage is confirmed; priority is released and normal signal operation resumes safely.
7	System records events, exceptions and metrics for later review.

5.2 Role-based access control

- Tier 1 — Traffic / government command: citywide operational view and authorized signal oversight.
- Tier 2 — Hospital emergency department: assigned incoming cases and approved patient information.
- Tier 3 — Ambulance operator: route guidance, request status and authorized destination updates.
- Administrative access should be separately controlled, monitored, and limited to necessary functions.

6. TRAFFIC PRE-EMPTION AND GREEN CORRIDOR

The Green Corridor engine estimates an ambulance's approach to an intersection and requests priority through an approved controller interface. The source concept describes an early approach around 1,000 metres, a nearer approach around 500 metres, and a release stage after passage. These are illustrative thresholds; actual timing must be designed for the road geometry, traffic controller and approved safety policy.

Phase	System action	Safety condition
A — Early approach	Identify route and upcoming junction; notify the controller or operator.	Validate location and request; do not assume signal state changed.
B — Near approach	Request or maintain priority according to controller status and policy.	Preserve pedestrian clearance and safe intergreen intervals.
C — Passage / release	Confirm passage, cancel or expire request, and return to normal timing.	Use controller-approved transition; log the result.

6.1 Multi-ambulance handling

When multiple ambulances approach a junction, the system must avoid conflicting signal phases. A scheduling policy may consider authorized dispatch priority, estimated arrival time, direction, and safety constraints. High-impact conflicts should be escalated to a traffic operator when necessary, and the reason for each decision should be recorded.

6.2 Non-signalized intersections

At selected junctions without traffic signals, roadside beacons may provide visual or audible warnings. Beacons require site assessment, suitable visibility, power and maintenance planning, and must supplement—not replace—safe driving and local traffic rules.



Figure 2. Example visual reference for an ambulance approaching a managed green corridor.

7. HOSPITAL INTEGRATION

Hospital integration aims to turn travel time into preparation time. An emergency department can receive an authenticated pre-alert with an estimated arrival time and only the information approved for the case. The dashboard supports staff coordination; it does not diagnose a patient or replace clinical judgement.

Step	Action	Output
1. Destination	Driver or dispatch selects a hospital under emergency-service procedures.	Destination recorded.
2. ETA update	Route service updates estimated arrival as new location data arrives.	Timestamped ETA and status.
3. Pre-alert	System sends approved case details to the receiving hospital.	Dashboard notification.
4. Preparation	ER team prepares personnel and resources according to hospital policy.	Acknowledgement or escalation.
5. Destination change	Authorized user updates the receiving facility.	Old alert reconciled; new facility notified.
6. Arrival	Arrival is confirmed and the case is closed.	Audit event and outcome record.

7.1 Data quality and privacy

- Display timestamps and source for vital signs or clinical details.
- Clearly flag missing, delayed or low-confidence information.
- Limit access to the receiving facility and staff with a valid care-related need.
- Use manual pre-alert procedures if the integration is unavailable.

7.2 Hospital matching

The nearest hospital may not always be the clinically appropriate destination. Matching should consider capability, reliable availability information, travel time, dispatch policy and patient circumstances. Destination decisions must follow the applicable emergency-service protocol and clinical judgement.

8. EMERGENCY IDENTITY TRACING

The project concept includes a workflow to help responders locate emergency contacts when a patient is unconscious or unidentified. This capability must be designed carefully: an ordinary emergency call does not automatically give an application access to a locked phone, SIM identifiers, IMEI data, carrier metadata or recent call records.

8.1 Proposed authorized workflow

Step	Action
1	Paramedic records that the patient is unidentified or unable to communicate.
2	System verifies the requester’s role, emergency purpose and authorization.
3	An approved registry or partner interface is queried only if available and legally permitted.
4	Potential contact information is checked according to the approved procedure.
5	A notification is sent through an approved channel, with minimum necessary disclosure.
6	The request, source, outcome and notification status are logged.

8.2 Privacy safeguards

- Use emergency contact details voluntarily supplied or held in an authorized registry, where available.
- Do not assume that a caller ID hash proves identity or a family relationship.
- Do not enable unrestricted access to carrier records or personal phone data.
- If lookup is unavailable or unauthorized, use the established manual procedure.
- Emergency stabilization and clinical decisions must not be delayed by the software’s inability to find a contact.

8.3 Community awareness

Public awareness materials may encourage people to carry a physical emergency card or contact sticker. Any QR card should disclose only carefully selected information, explain who can access it, and provide a non-digital alternative.

9. SECURITY AND AUTHORIZATION

Because the system may handle location, emergency-case, health and contact data—and may request traffic-signal actions—security must be built into the architecture.

Control	Purpose
Strong authentication	Verify users and service identities before granting access.
Role-based permissions	Limit each user to functions and records required for their role.
Signed, short-lived requests	Reduce the risk of forged or replayed sensitive commands.
Encryption	Protect data in transit and appropriately protect stored sensitive information.
Audit logs	Record who initiated an action, when, its purpose and its outcome.
Monitoring and alerts	Identify service outages, unusual access and repeated failed requests.
Safe fallback	Keep traffic controllers in their approved safe mode if cloud coordination fails.
Incident response	Define actions for compromised accounts, incorrect matches, outages and unauthorized access.

9.1 TOTP and zero-trust design

A time-based one-time password (TOTP) can be one login factor, but it is not sufficient by itself to authenticate every vehicle or make a signal command safe. The design should also validate the device or service, the request, its expiry, the user’s permissions and the controller’s acknowledgement.

9.2 Governance

Live traffic control, hospital-system integration, and any telecom or identity lookup require formal authorization and review by the responsible authorities and qualified legal, cybersecurity, privacy and clinical stakeholders in India.

10. IMPLEMENTATION PLAN

Phase	Activities	Deliverable
Phase 1: Planning and design	Requirements, stakeholder discussions, workflow mapping, architecture and risk review.	Approved requirements and pilot boundary.
Phase 2: Prototype development	Mobile app, dashboards, route/ETA service, simulated signals and mock hospital integration.	End-to-end software prototype.
Phase 3: Testing	Functional, integration, security, usability, performance and failure tests.	Test report and critical issues resolved.
Phase 4: Shadow mode	System generates recommendations while existing operators remain in control.	Baseline comparison and user feedback.
Phase 5: Limited pilot	Supervised operation at approved sites with fallback and rollback plans.	Pilot results and safety review.
Phase 6: Scale decision	Review cost, measured benefit, risks, support needs and partner readiness.	Decision to expand, revise or stop.

10.1 Proposed project team

- Project lead / product owner — scope, coordination and reporting.
- Backend and integration engineer — APIs, authentication and service logic.
- Mobile and frontend developers — ambulance app and dashboards.
- Traffic systems engineer — signal integration and safety constraints.
- Healthcare workflow adviser — hospital pre-alert and clinical data requirements.
- Cybersecurity / privacy specialist — threat model, governance and security testing.
- QA / test engineer — functional, failure and acceptance testing.

10.2 Prototype demonstration checklist

- Simulated ambulance moves along a route and triggers a request.
- Traffic-priority request shows pending, acknowledged, rejected or timed-out states.
- Hospital dashboard receives an ETA and approved mock data.
- Identity/contact workflow uses synthetic data and authorization checks.
- System records audit events and demonstrates safe fallback when a service fails.

11. TESTING AND EVALUATION

Test area	Example check	Expected result
Authentication	Expired credentials attempt a sensitive action.	Request rejected and logged.
Location quality	GPS update is stale or inaccurate.	System flags uncertainty and avoids unsafe assumptions.
Traffic interface	Controller does not acknowledge a request.	No false success; operator alert and fallback.
Multi-ambulance	Two vehicles approach conflicting directions.	No conflicting signal commands.
Hospital alert	Pre-alert is delayed or destination changes.	Status is updated and stale alerts handled.
Contact lookup	Request has no valid authorization.	Lookup blocked; manual care process remains available.
Outage	Network or cloud service becomes unavailable.	Safe degraded mode and traceable recovery.

11.1 Metrics to collect

- Ambulance transit time and measured signal delay, where available.
- ETA error compared with actual arrival time.
- Hospital pre-alert delivery and acknowledgement time.
- Contact workflow success rate and time to verified outcome.
- Signal request acceptance, rejection, timeout and cancellation rates.
- System uptime, incident count, recovery time and safety events.

12. EXPECTED OUTCOMES

The original concept proposes a 45% reduction in transit time, 60% faster ER triage and 80% faster identity resolution. These figures are targets to investigate—not verified results. They must be evaluated using baseline measurements, a defined pilot, consistent definitions and transparent reporting.

Proposed target	How to validate it
45% lower transit time	Compare baseline and pilot travel times on comparable routes and traffic conditions.
60% faster ER triage	Use hospital-approved timestamps and account for case mix and staffing.
80% faster identity resolution	Measure authorized lookup time, including unsuccessful cases and manual escalation.
Improved security	Measure access-control test results, audit completeness, incidents and unresolved risks; do not promise 100% security.

13. RISK ANALYSIS AND MITIGATION

Risk	Possible impact	Mitigation
GPS/network failure	Wrong ETA or missing updates.	Location-quality checks, monitoring and fallback.
Unauthorized signal request	Traffic disruption or safety hazard.	Strong authentication, signed requests, controller interlocks and audit.
Multiple ambulances	Conflicting priorities.	Explicit scheduling policy and operator escalation.
Incorrect contact match	Privacy harm or wrong notification.	Approved data sources, verification and minimum disclosure.
Hospital integration failure	Late or stale pre-alert.	Manual fallback and timestamped status.
Overstated results	Poor decisions or loss of credibility.	Clearly label targets and publish measured evidence.
Maintenance gaps	Reduced reliability over time.	Named owners, monitoring, training and service plans.

14. CONCLUSION AND FUTURE SCOPE

The Smart Emergency Green Corridor System proposes a coordinated approach to urban emergency response by linking ambulance route information, safe traffic-signal priority, hospital pre-arrival preparation, and authorized emergency-contact assistance. Its central value is improved coordination between systems and teams that may otherwise operate separately.

A safe implementation should begin with a simulation prototype, proceed to shadow-mode evaluation, and move to live traffic only after formal authorization, testing, safety review, and rollback planning. Identity tracing must use approved data sources and must never assume unrestricted access to locked phones or private carrier records.

Future enhancements may include improved congestion prediction, approved blood-bank availability interfaces, authorized health-record integration, and aggregated operational analytics. Each extension requires separate feasibility, privacy, and safety review.

15. REFERENCES AND DESIGN NOTES

- Primary source: user-provided “Project Golden Hour Implementation Report” (10 pages).
- Consult current official traffic-controller documentation and local traffic-authority procedures before integration.
- Confirm applicable Indian data-protection, telecommunications and healthcare requirements with qualified advisors.
- Use approved hospital procedures and interoperability specifications for any clinical-data exchange.
- Treat all projected impact values as hypotheses until supported by measured pilot results.

SMART EMERGENCY GREEN CORRIDOR SYSTEM

Faster response • Safer coordination • Evidence-based emergency care